

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale(short)	Comments
FR-001	Functional	The system shall match the driver's vehicle connector type (CCS, CHAdeMO, Type 2) with available station chargers and reserve a 30 minute charging window	High	Pass: incompatible connector plugs hidden from driver. Fail: two vehicles booked for same socket concurrently	Ensures compatibility and prevents double-booking.	
FR-002	Functional	The system shall allow the driver to search for charging stations within a selected radius of their GPS location, filtered by connector type.	High	Setting a 5 km radius + connector filter returns only compatible stations within that radius.	Drivers need to quickly find reachable, compatible stations.	
FR-003	Functional	The system shall calculate and display the estimated dynamic charging cost before the driver confirms a reservation.	High	Estimated cost (rate × expected duration) appears before the "confirm" button is enabled.	Transparent pricing supports informed decisions.	
	Functional	The system shall allow the driver to cancel a reservation up to 15 minutes before slot start without penalty.	Medium	Cancelling at T-16 min or earlier shows "no charge"; at T-14 min or later, a penalty applies.	Gives flexibility while protecting near-term utilization.	
	functional	The system shall allow the station operator to update a charger's status (Available/Occupied/Out-of-Service) in real time.	medium	Marking a charger "Out-of-Service" removes it from driver search results within 5 seconds.	Keeps availability data accurate, prevents failed reservations.	

NF R- 001	Non- Functional (performance and security)	The system shall automatically release a locked socket and apply a penalty if the driver fails to plug in within 10 minutes of slot start.	High	Benchmark tests confirm target latency and security standards under simulated peak load.	Prevents socket hoarding, keeps chargers free for others.	
NF R- 002	Non- Functional (performance and scalability)	The system shall support at least 500 concurrent reservation requests during peak hours with average response time under 2 seconds.	Hight	Load test with 500 concurrent users shows avg. response < 2s and 0 dropped requests.	Keeps system responsive during high demand (holidays, rush hours).	